

Srivatsan Balaji

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EDUCATION

University of Michigan

Ann Arbor, MI

B.S.E in Computer Engineering, Minor in Mathematics (GPA: 3.9/4.0)

May 2028

- **Relevant Awards:** BPA C++ Programming – 2nd Place (Regional), BPA Financial Analyst Team Captain – 1st Place (Regional), HOSA College Physics – 2nd Place (State), BPA National Torch Leadership Award
- **Relevant Coursework:** Data Structures & Algorithms, Digital Logic Design, Discrete Math, Differential Equations, Linear Algebra, Multivariable Calculus

WORK EXPERIENCE

Firmware Engineering Intern (Incoming)

May. 2026 – Aug. 2026

Trackonomy Systems, San Jose, CA

- Develop embedded firmware in C/C++ for IoT asset-tracking devices supporting large enterprise shipping networks
- Collaborate with hardware teams to debug device communication, optimize power usage, and improve reliability
- Integrate wireless communication modules, including BLE and LoRa, with ARM-based microcontrollers and onboard sensors to enable low-power, long-range tracking

Embedded Firmware Researcher

Jan. 2026 – Present

University of Michigan Atombot Lab, Ann Arbor, MI

- Optimize embedded firmware on a Raspberry Pi through a HAL to reduce real-time latency in robotics applications
- Deploy and validate FreeRTOS code, electronic components, and NFC interfaces, performing system level testing
- Develop firmware and CAN control algorithms in C/C++ for robotic systems, enabling wheel-legged kinematics

High Voltage Embedded Software Developer

Sep. 2025 – Jan. 2026

SPARK Electric Racing, Ann Arbor, MI

- Assist with testing and validation of high-voltage embedded control systems, using Altium for schematic and PCB tests
- Develop and integrate microcontroller firmware in C for real-time communication with BMS modules over CAN bus
- Engineered an integrated isolation board, improving system safety, robustness, and fault tolerance for circuit leaks

Software Development Intern

Jul. 2025 – Aug. 2025

ComputeK Solutions, Novi, MI

- Developed C++ programs to automate financial modeling (FCFE, EBITDA, DCF, etc.), reducing analysis time by 70%
- Built reusable valuation libraries to document and streamline business analysis for future equity-based transactions
- Research market trends, analyze comparable companies and precedent transaction analysis, etc. to support VC deals.

PROJECTS

Low-Power Embedded Sensor Logger (Ongoing)

- Implementing embedded C/C++ firmware on STM32 microcontroller to log accelerometer and temperature data seeking <5% CPU utilization
- Optimizing low-power operation using deep sleep modes, interrupt-driven firmware, and peripheral clock gating
- Integrating SD card storage and UART/Bluetooth communication for real-time data logging and offline analysis

Embedded Bluetooth Hardware

- Developing a Bluetooth transmitter module to interface with a SONY NWE-505 MP3 player
- Designed custom PCB layout and programmed embedded microcontroller firmware to enable wireless audio streaming
- Implemented a low-power communication system to optimize in-module battery efficiency and signal stability

Real-Time Audio Spectrum Visualizer

- Developed FFT-based audio signal processing in embedded C on STM32 to visualize frequency spectrum on OLED display
- Reduced latency by 30% through fixed-point arithmetic and optimized buffer management on constrained hardware
- Designed visualization system UI demonstrating real-time DSP capabilities on STM32 microcontroller platform

SKILLS

Programming Languages: C++, C, Python, Java, JavaScript, Verilog

Hardware & Tools: ESP32, STM32, Communication Protocols (I2C, UART, CAN), FPGA, Altium, Git, Jira